

BANARAS HINDU UNIVERSITY

B.Sc. — Semester IV Mid-Semester Examination, 2025-26 (NEP)

MATHEMATICS**Paper Code: MAT-MJ-42 : Vector and Tensor Analysis (Mid-Semester)**

Time: 1 Hour

Maximum Marks: 20

*Instructions: Attempt any **FOUR** questions. The figures in the right-hand margin indicate marks.*

Question 1. Give the definition of curl of a vector function. If $\vec{A}$ and $\vec{B}$ are two vector functions, then prove that

$$\text{curl}(\vec{A} \times \vec{B}) = (\vec{B} \cdot \text{grad})\vec{A} - \vec{B}(\text{grad} \cdot \vec{A}) - (\vec{A} \cdot \text{grad})\vec{B} + \vec{A}(\text{grad} \cdot \vec{B}).$$

[5]

Question 2. Give the definition of gradient of a scalar function. Find the equation of tangent plane to the surface $4x^2z + yz^2 + xy^2 = 2$ at the point $(1, 2, -1)$.

[5]

Question 3. Prove that

$$(i) \text{curl}(\text{grad } \phi) = 0,$$

$$(ii) \text{div}(\text{curl } \vec{A}) = 0,$$

where ϕ is a scalar function and $\vec{A}$ is a vector function.

[5]

Question 4. Evaluate

$$\int_c (2xz^2 dx + 3z^2 dy - 3x^2y dz),$$

where c is the curve given by $x = t^3$, $y = 2t$, $z = t^2$, from $t = 0$ to $t = 2$.

[5]

Question 5. Evaluate

$$\iint_S \vec{F} \cdot \hat{n} ds,$$

where $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and S is the surface of the cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$.

[5]

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